

Globe

PROJECT FOR THE “INGEGNERIA DEL SOFTWARE E PROGETTAZIONE WEB” COURSE

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1. Software Requirement Specification

1.1. Introduction

1.1.1. Aim of the document

The main point of the following document is present objectively the fundamental requirements and specific ones of the system *Globe*. So, we will be doing a review of all the important internal and external aspects of the system with the help of some diagrams.

1.1.2. Overview

Globe is an application designed to facilitate the creation of travel itineraries.

User can request itineraries to travel agency showing what he wants to visit and view agency proposal that he can accept, paying proposal's price, or reject. Moreover, user can generate itineraries by selecting cities and attractions he wants to visit, adding accommodation and flight departure and arrival time (this function is available for guest users too – demo mode).

Agency can view user requests and create a proposal to send to the user and view if its proposals have been accepted or not.

1.1.3. HW e SW requirements

The hardware requirements are:

- A computer or a laptop
- An internet connection for city and attraction search
- Some space in the disc to save files if the demo version is on.

The software requirements are:

- Oracle OpenJDK 22 correctly installed
- A server or a local server that can holds the DBMS to obtain information by queries in SQL

1.1.4. Related systems Pros and Cons.

- Sygic Travel
 - Pros: it offers additional features such as booking tour packages, sharing itineraries and the multi-device account access support.
 - Cons: users must create a new itinerary for every city when planning multi-city trips and it includes a smaller number of attractions.
- SiVola
 - Pros: it includes an intuitive and user-friendly interface and offers the possibility to purchase a pre-tested itinerary from the agency for a unique experience.
 - Cons: it doesn't offer the option to create itineraries independently or to request a custom-made itinerary.

1.2. User Stories

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- As a user, I want to get the shortest route, so that I can visit all the attractions on my itinerary.
- As an agency, I want to see all attractions and cities user want to visit, so that I can create a complete itinerary.
- As a user, I want to see the details of the agency's proposal and the related itinerary, so that I can decide to accept or reject the proposal.

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- As a user, I want to request an itinerary indicating its type, so that the agency can create the best itinerary for me.
- As an agency, I want to see the history and status of the proposal made to the user, so that the agency offers a more efficient service to the costumers.
- As a user, I want to rate the agency after payment, so that I can help other users choose the best agency.

1.3. Functional Requirements

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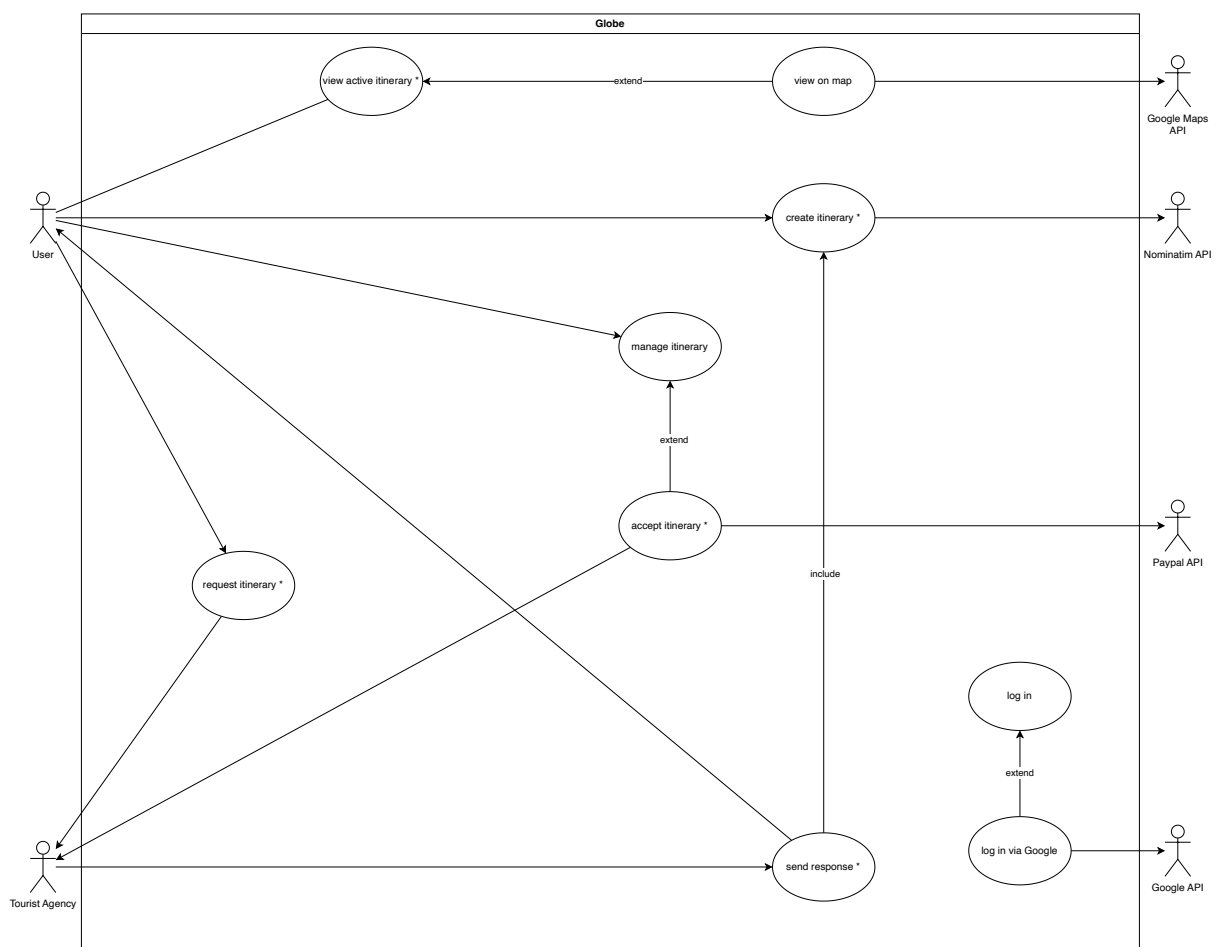
- The system shall provide a real-time updated map to see the user's active itinerary.
- The system shall find the available agencies according to the itinerary's types requested by the user.
- The system shall calculate the shortest route between the attractions selected by the user.

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- The system shall provide a secure payment method for the user to use.
- The system shall save and display to the user the proposal's itinerary after payment is completed.
- The system shall notify users when their request is accepted by an agency.

1.4. Use Cases

1.4.1. Overview



1.4.2. Internal Steps

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Name: Create Itinerary

1. The user Logs In.
2. The user fills the creation form with itinerary details
3. The user searches cities and attractions he wants to add.
4. The system finds cities and attractions from the API.
5. The user selects correct cities and attractions.
6. The user submits the form to confirm the itinerary.
7. The system verifies itinerary's data.
8. The system calculates the optimal route.
9. The System saves the itinerary.

Extensions:

- 4a. CITY OR ATTRACTION NOT FOUND
System notifies the user about the city or the attraction it doesn't found.
- 4b. API NOT RESPONDING
System notifies the user about the problem.
- 7a. DATA NOT VALID
System notifies the user about the incorrect field.

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Name: Request itinerary

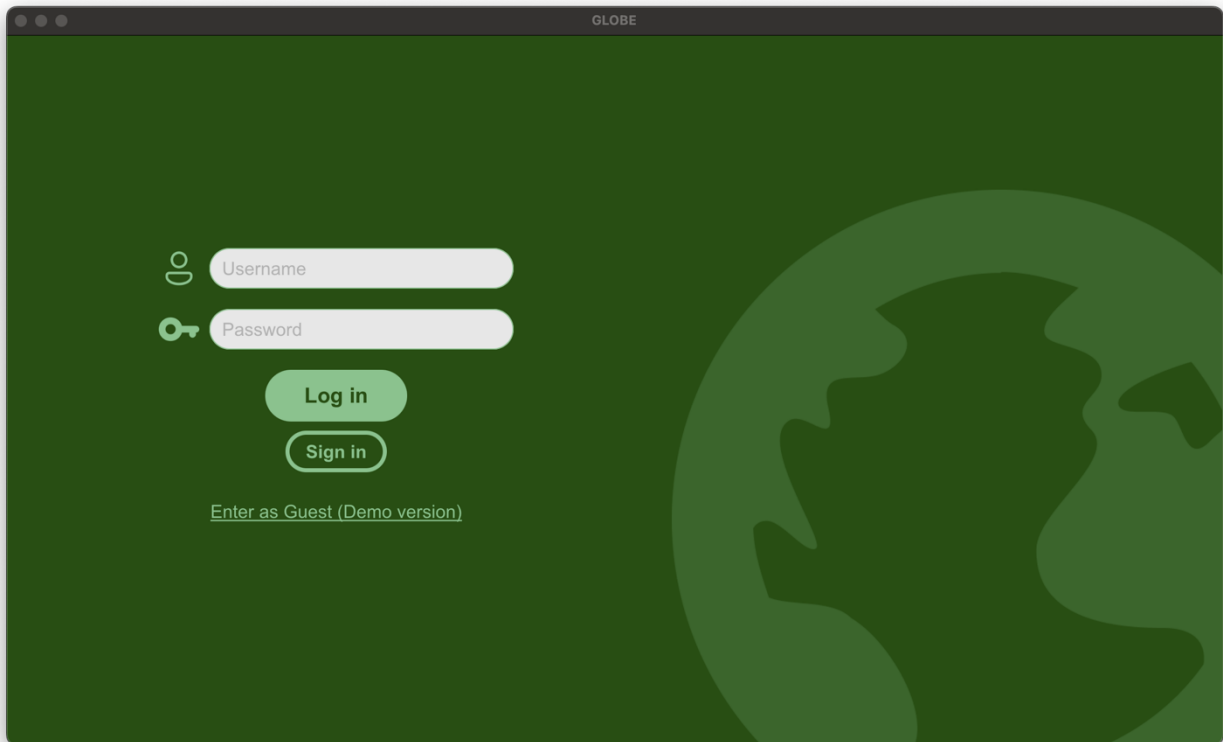
1. The user Logs In.
2. The user searches for cities and attractions to include in the itinerary request.
3. The system retrieves matching cities and attractions from the API.
4. The user selects the desired cities and attractions from the results and compile the request form.
5. The system validates the data in the request.
6. The user searches for agencies.
7. The system finds the agencies based on the entered criteria.
8. The user selects the agencies to send the request to.
9. The system sends the validated request to the selected agencies.

Extensions:

- 5a. DATA NOT VALID
System notifies that the data entered is not valid.
- 7a. AGENCY NOT FOUND
System notifies the user that there aren't agencies available based on the user's input.
- 8a. NO AGENCY SELECTED
System notifies the user that he has not selected any agency.

2. Storyboards

Log In:



Manage Itinerary:



Create Itinerary:

The 'Create Itinerary' form in the GLOBE application features a dark green background with a large globe illustration. The form includes the following elements:

- Title:** "Weekend in London" with a "Day: 2" selector.
- Description:** "Two days in the capital of Great Britain, an itinerary to see the main attractions in the shortest time possible."
- Category Selection:** "The Road" (selected), "Nature", "Culture", "Relax", and "City".
- City:** "City Name" field with "London" entered.
- Attractions:** "Attraction Name" field with a list of suggestions: "Big Ben", "Westminster Park", "Camden Town", "Camden Road", "The Shard", and "London Eye".
- Accommodation:** "Address" and "Name" fields with "PubLove Hostel, 21th street, 12" entered.
- Flight:** "Inbound Flight" (12.00 - 14.30) and "Outbound Flight" (21.15 - 23.45) fields.
- Navigation:** "Back" and "Next" buttons.

The bottom navigation bar contains "Richiedi" (Home icon), "Trip" (Briefcase icon), and "Log Out" buttons.

Display Itinerary:

The 'Display Itinerary' view in the GLOBE application displays the created itinerary with the following details:

- Title:** "Weekend in London" with "2 Days" indicator.
- Description:** "Two days in the capital of Great Britain, an itinerary to see the main attractions in the shortest time possible."
- Category Selection:** "Culture" and "City" (selected).
- City:** "City: London, United Kingdom".
- Attractions:** A list of attractions: "Big Ben", "Camden Market", "Camden Town", "London, Anselm Road", "London, Camden Lock Place", and "London, Camden High Street".
- Accommodations:** "1: Pub Love Hostel, 21th street 12".
- Flight:** "Inbound: 12.0 - 14.2" and "Outbound: 21.3 - 23.5".
- Navigation:** "Back" and "Next" buttons.

The bottom navigation bar contains "Richiedi" (Home icon), "Trip" (Briefcase icon), and "Log Out" buttons.

Create Request:

The screenshot shows the 'Create Request' form in the GLOBE application. The form is set against a dark green background with a subtle globe pattern. It includes the following elements:

- Header:** The GLOBE logo and a title bar with the text 'GLOBE'.
- Search Fields:** Two input fields with placeholder text: 'Where you want to go?' and 'What you want to visit?'. Each field has a right-pointing arrow button.
- Other requests:** A horizontal bar with a label 'Other requests' and a right-pointing arrow button.
- Filters:** A row of buttons with plus signs and labels: '+ On The Road', '+ Nature', '+ Culture', '+ Relax', and a minus sign button.
- Optional Add-ons:** A section labeled 'Add optional:' with buttons for '+ Flight' and '+ Accommodation', followed by a 'How many days?' label and a numeric input field with a value of '0' and minus/plus buttons.
- Travel mode:** A dropdown menu is open, showing options: 'On The Road' (selected), 'Nature', 'Culture', and 'Relax'. Below the dropdown, there are three buttons for 'Travel mode': 'Morning', 'late afternoon', and 'Nigh'.
- Day driving hours:** A section labeled 'Day driving hours' with a text input field containing the placeholder 'How many hours do you want to drive a day?'.
- Buttons:** A 'Get Agencies' button with a right-pointing arrow and a 'Next' button with a right-pointing arrow.
- Footer:** A navigation bar with three buttons: a home icon labeled 'Richiedi', a suitcase icon labeled 'Trip', and a 'Log Out' button.

Accept Proposal:

The screenshot shows the 'Accept Proposal' screen in the GLOBE application. It features a dark green background with a globe pattern. The layout includes:

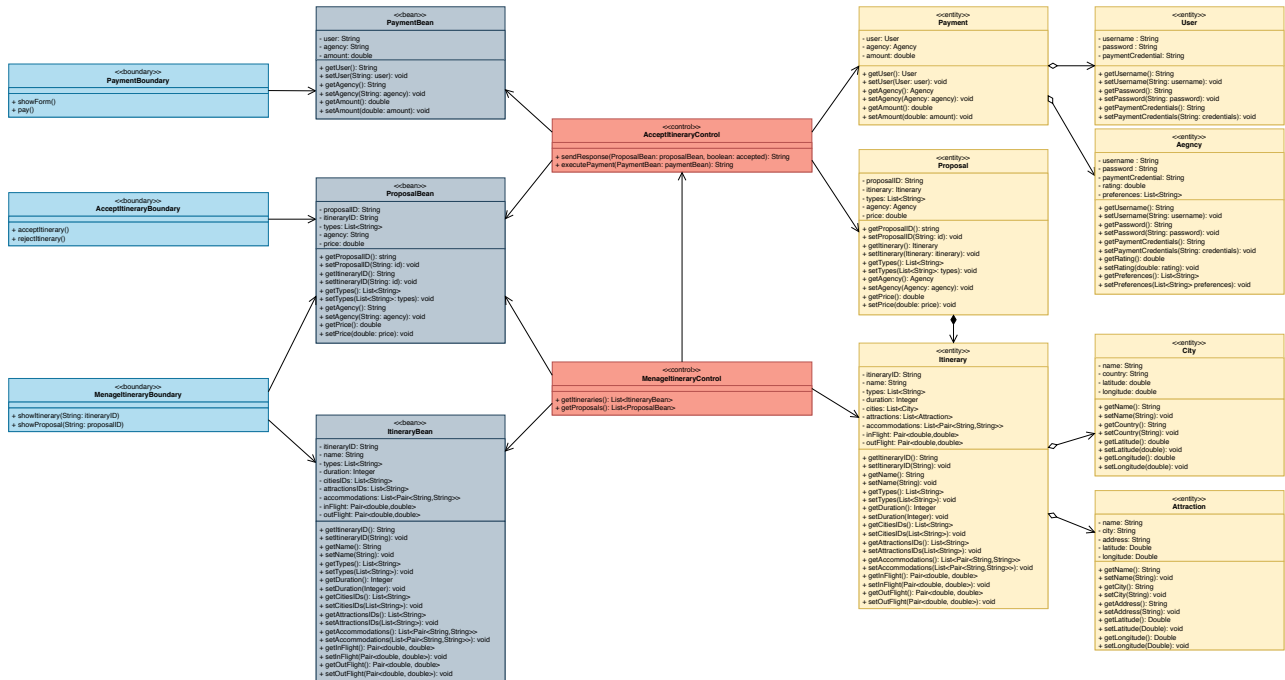
- Header:** The GLOBE logo and a title bar with the text 'GLOBE'.
- Navigation:** A 'Back' button with a left-pointing arrow.
- Proposal Details:** The name 'SiVola' is displayed, followed by the text 'One day in rome'. To the right, the 'Price' is listed as '289.5'.
- Actions:** An 'Itinerary' button with a right-pointing arrow, and two buttons at the bottom: 'Accept' and 'Reject'.
- Footer:** A navigation bar with three buttons: a home icon labeled 'Richiedi', a suitcase icon labeled 'Trip', and a 'Log Out' button.

3. Design

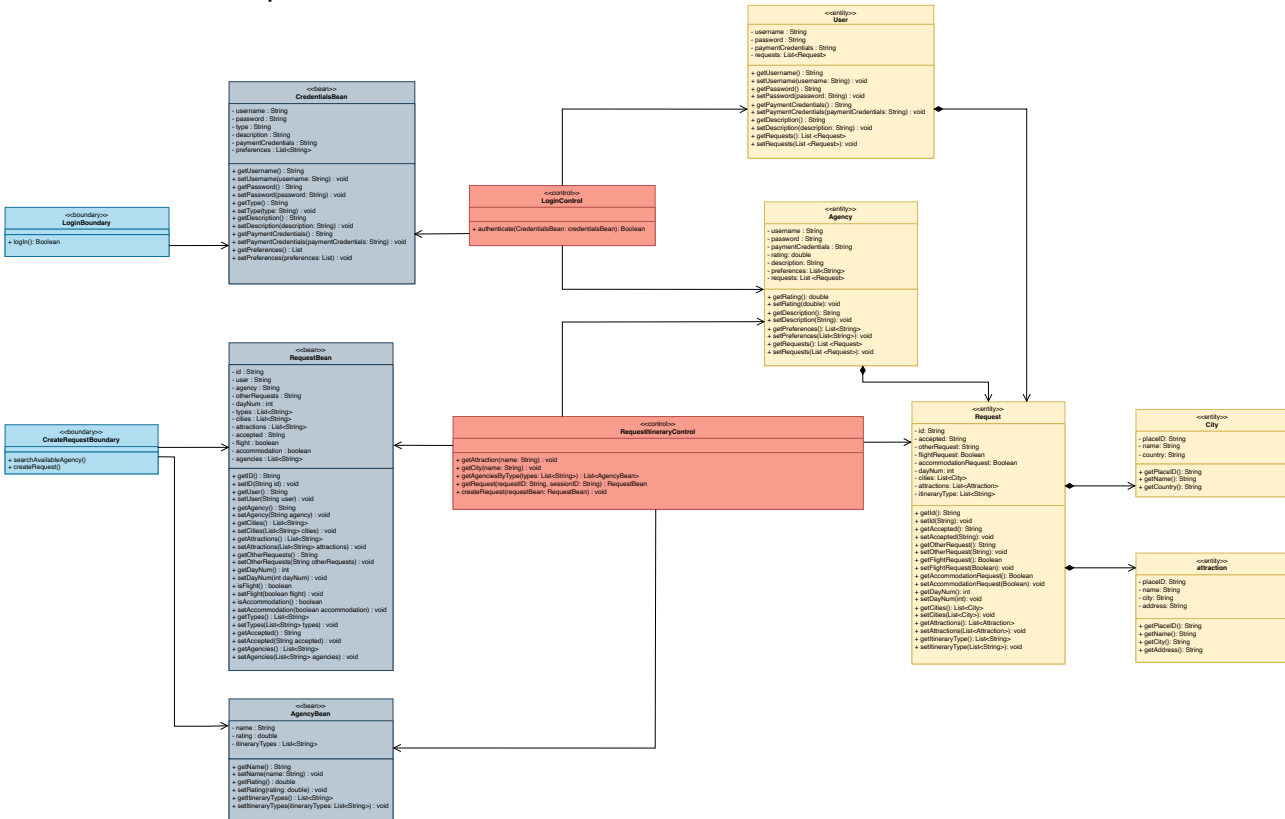
3.1. Class diagram

3.1.1. VOPC

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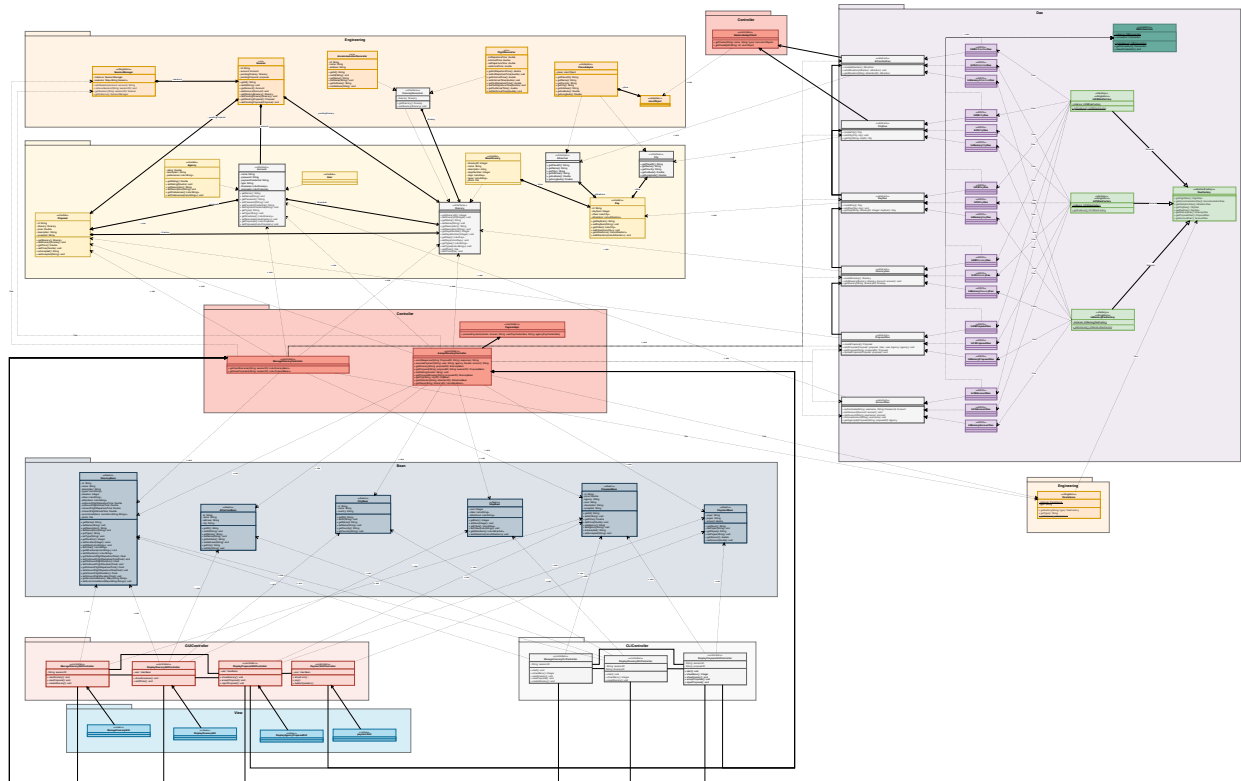
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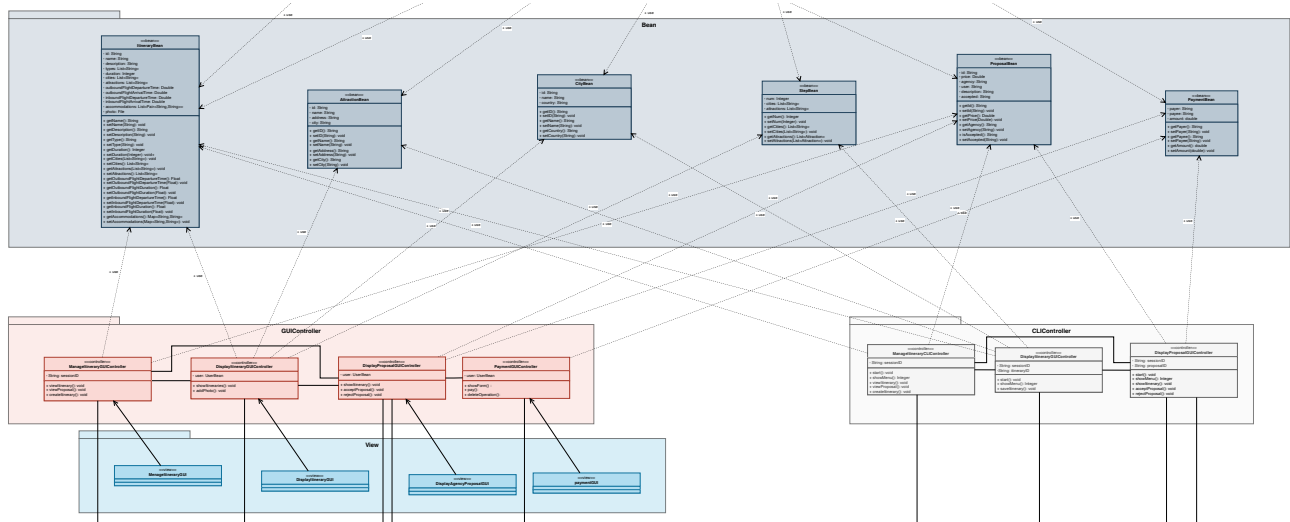
3.1.2. Design-level diagram

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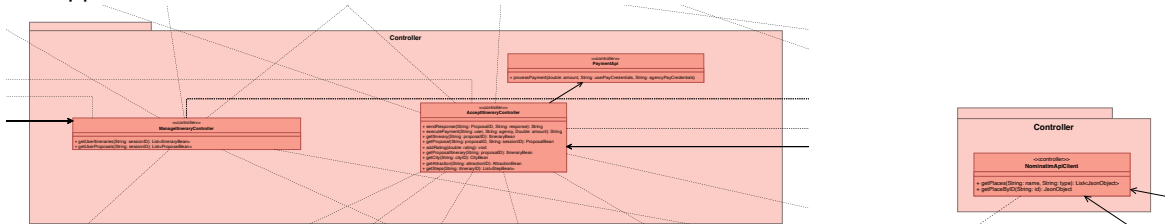
Complete Diagram:



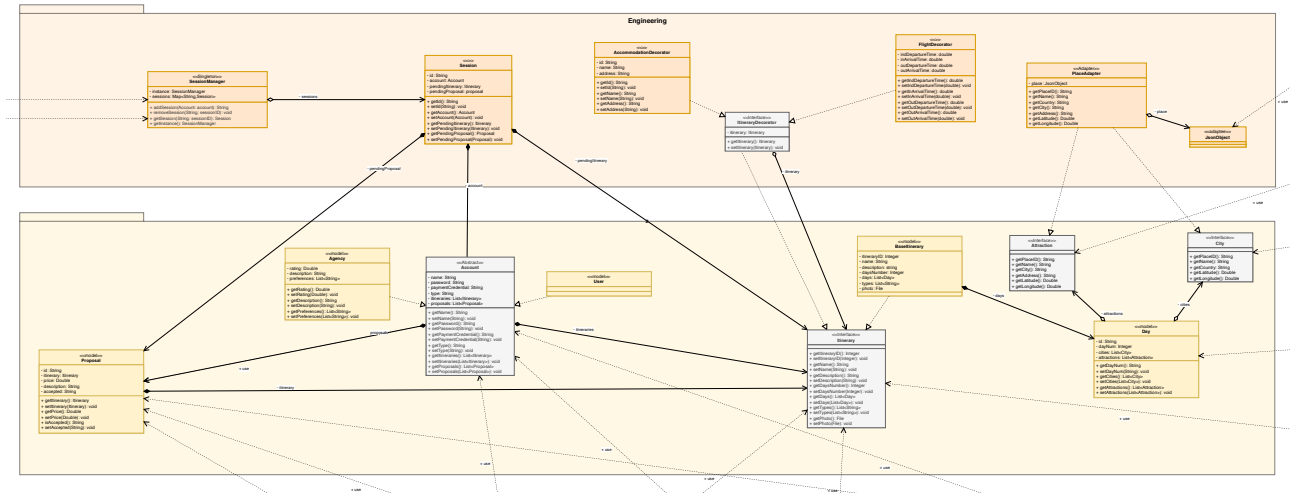
Graphic Controller and Bean:



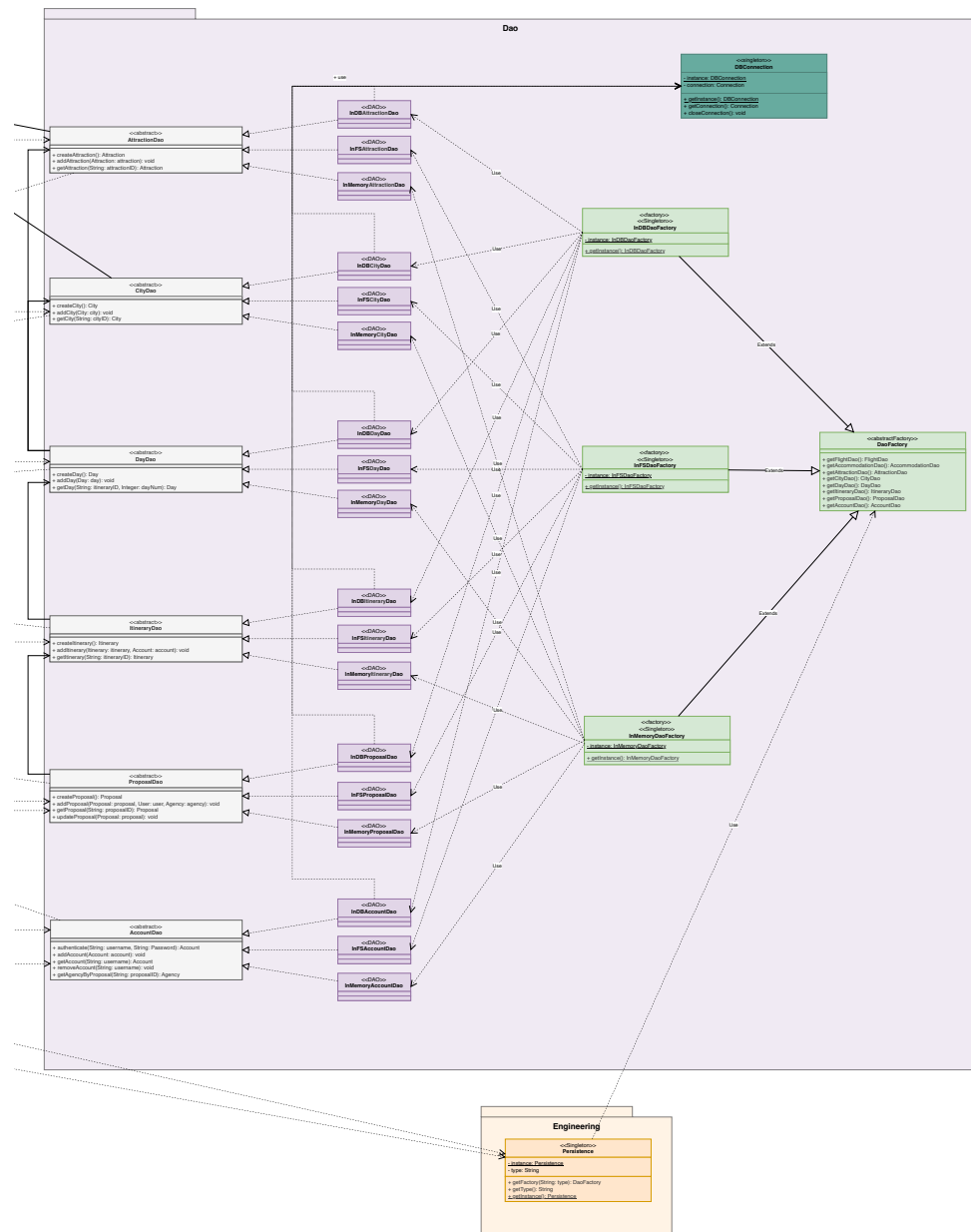
Application Controller:



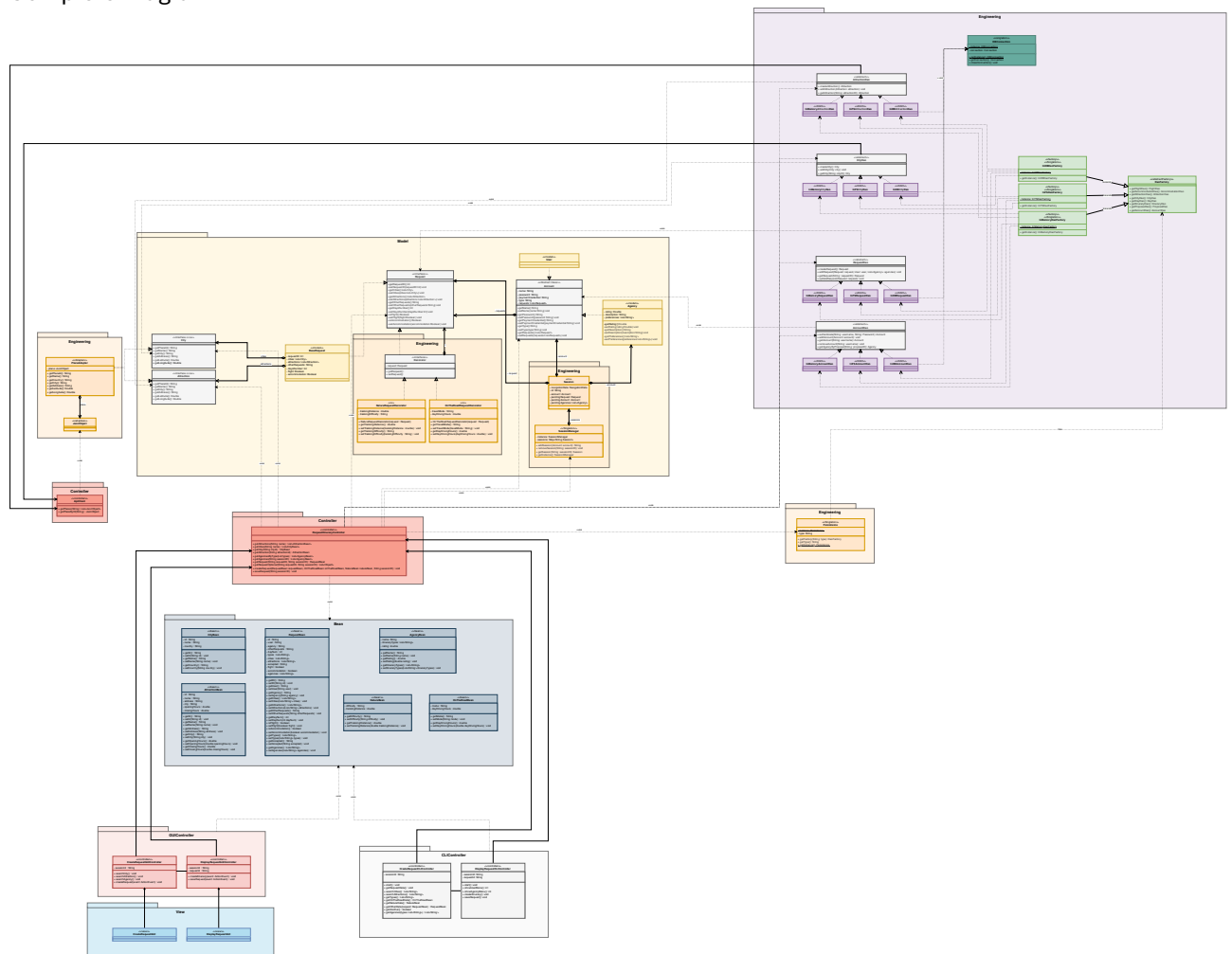
Model and Engineering:



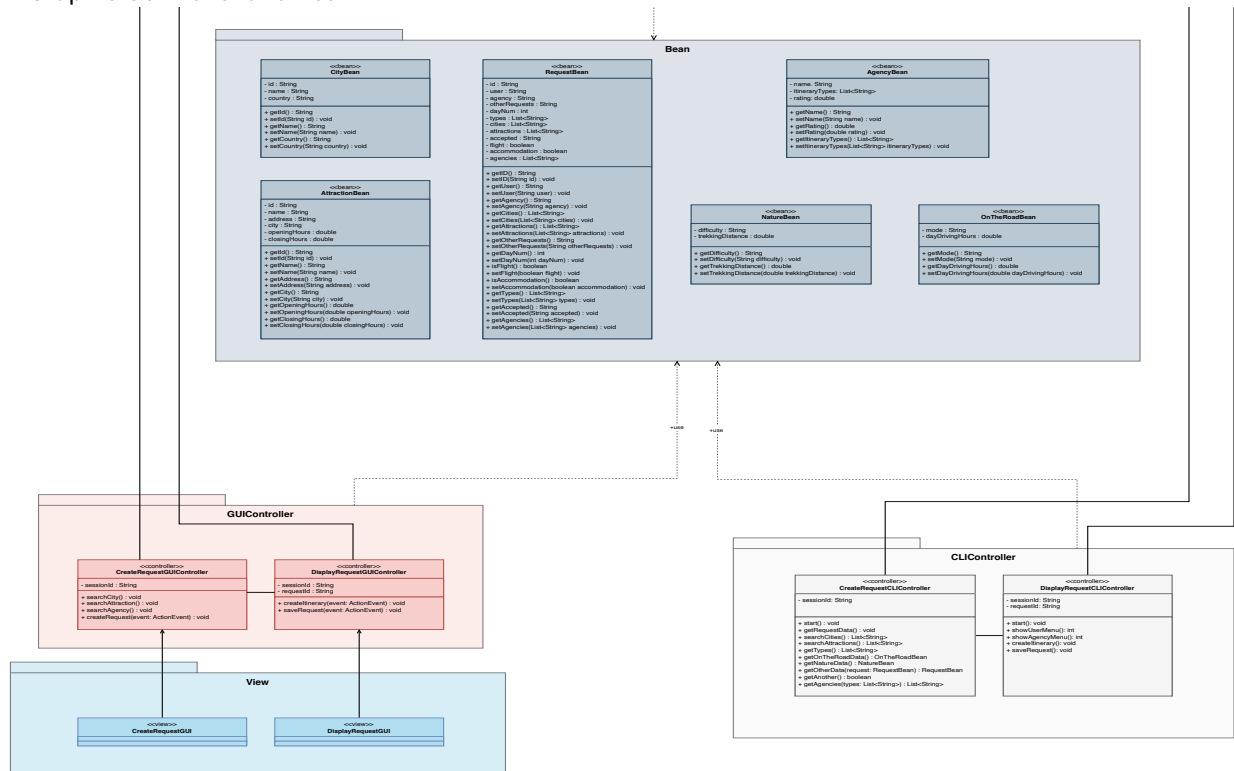
DAO:



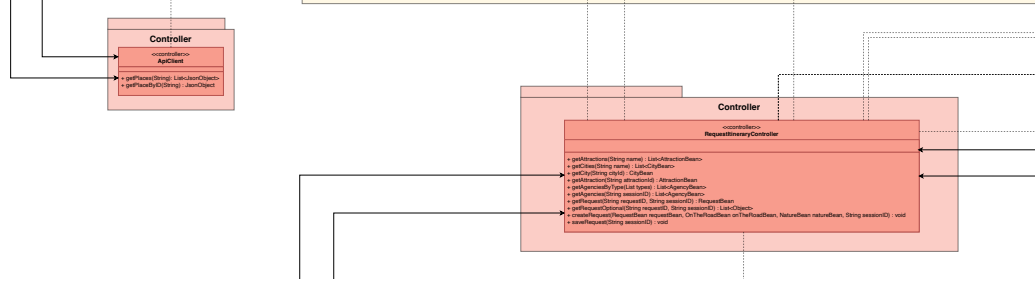
Simone Tummolo | 0309116: Complete Diagram:



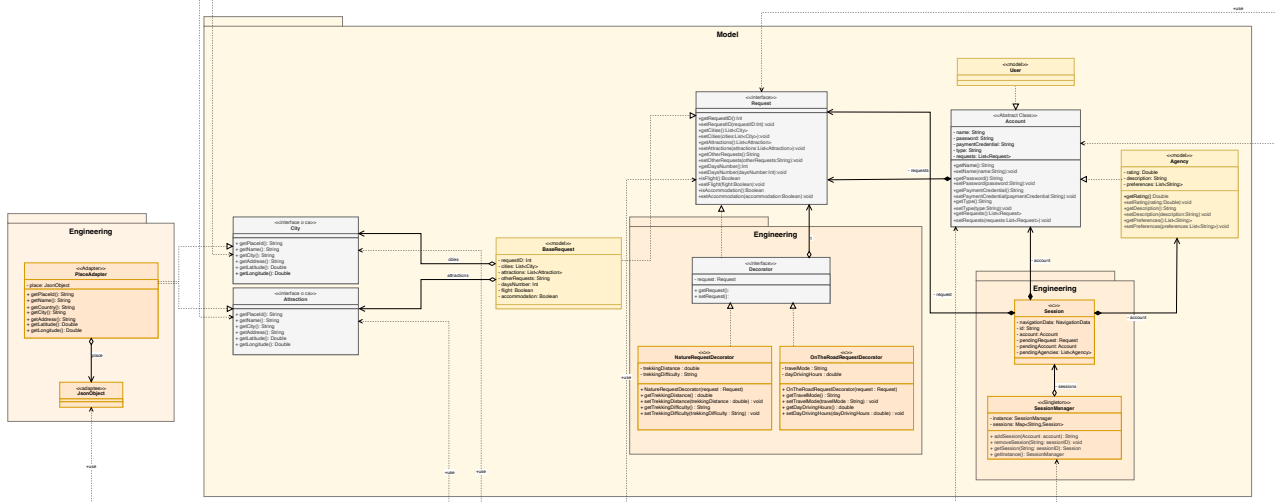
Graphic Controller and Bean:



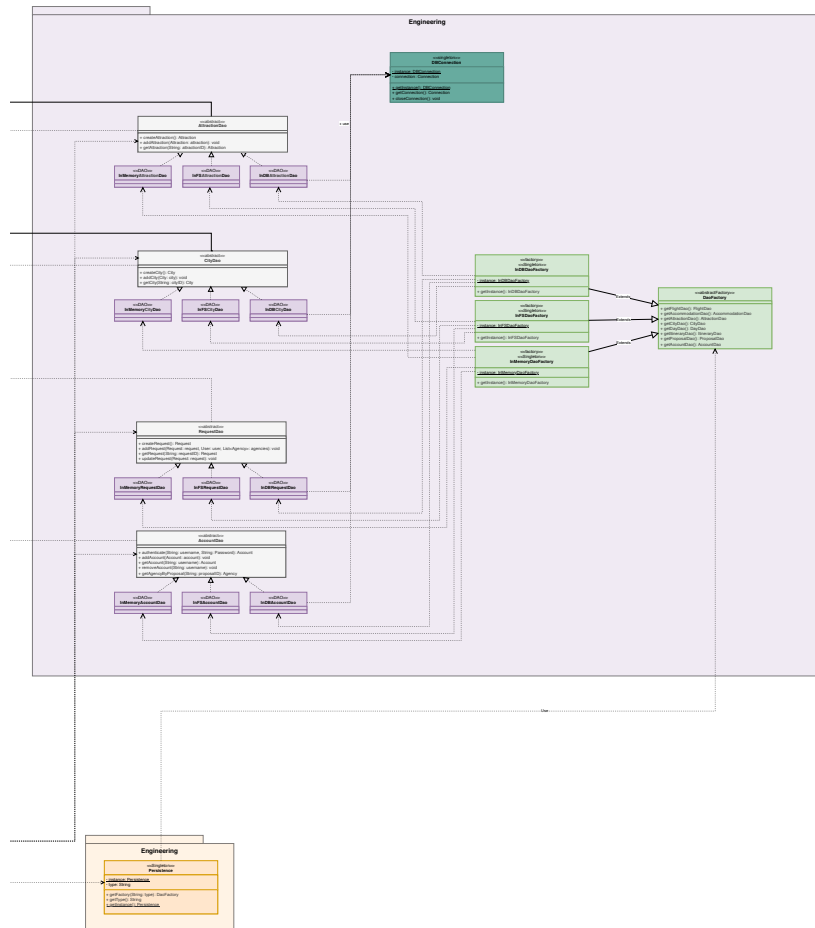
Application Controller:



Model and Engineering:

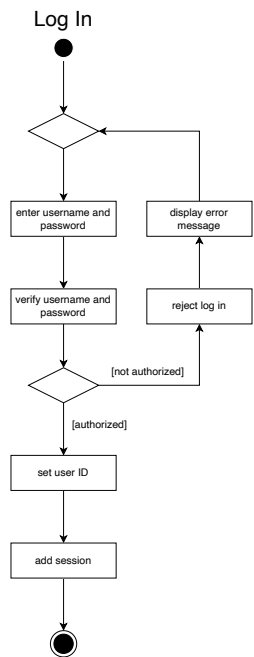
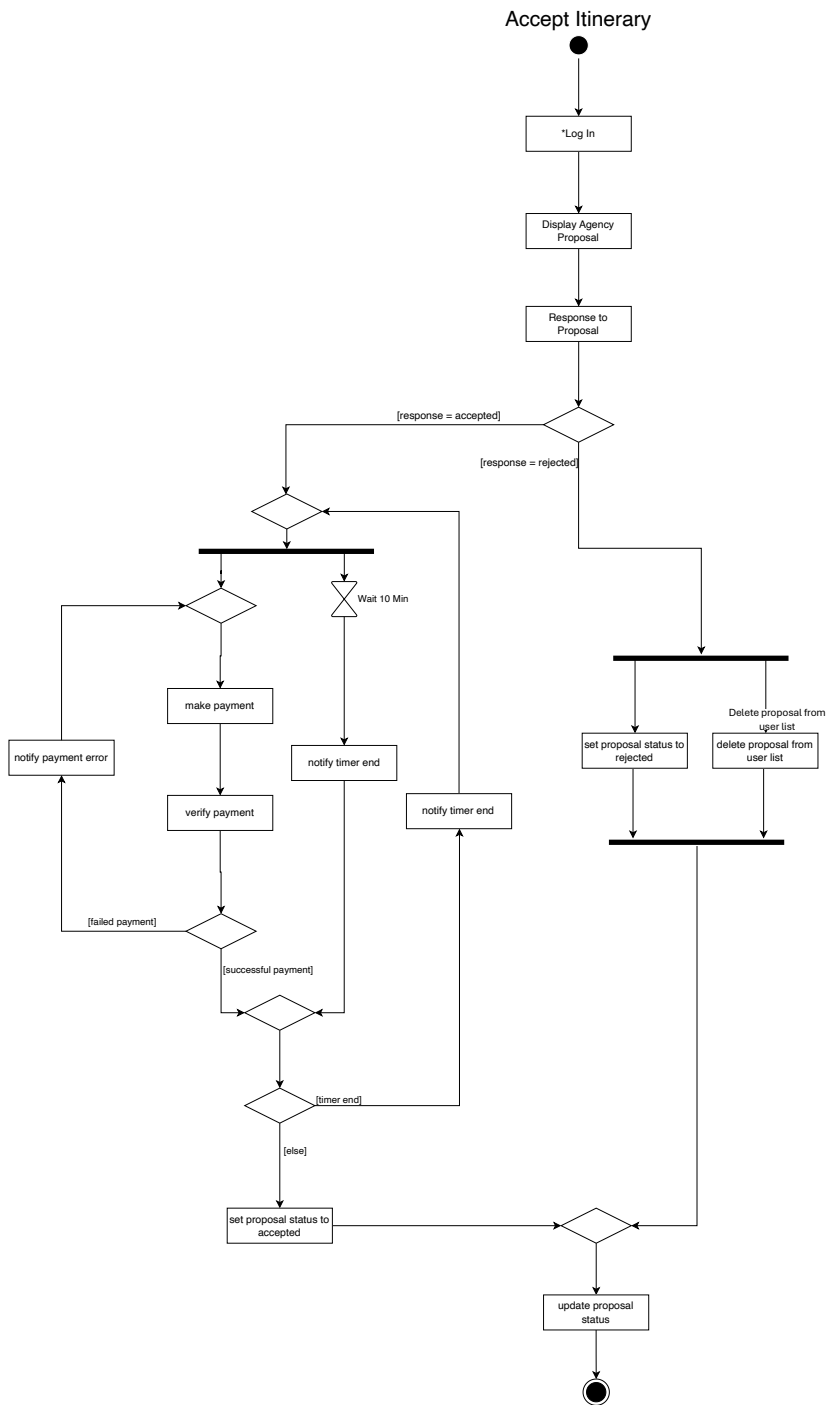


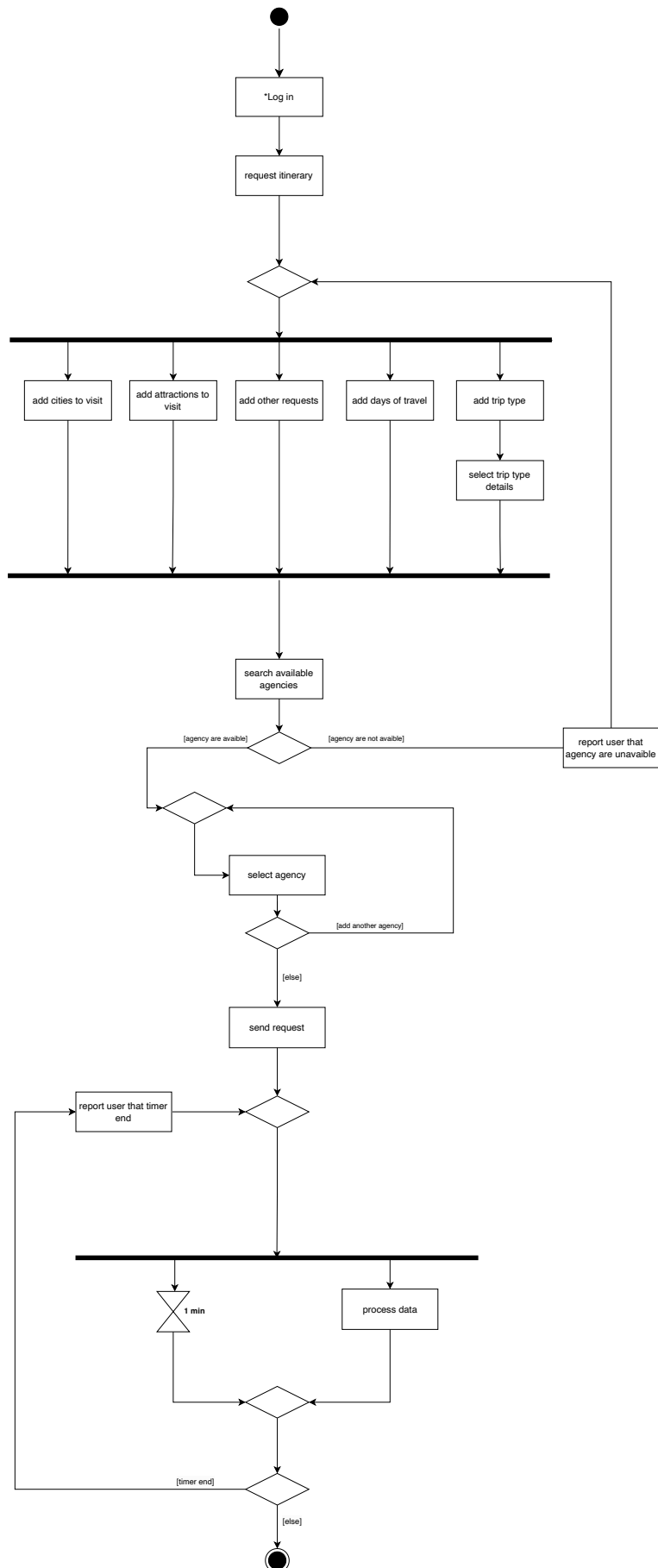
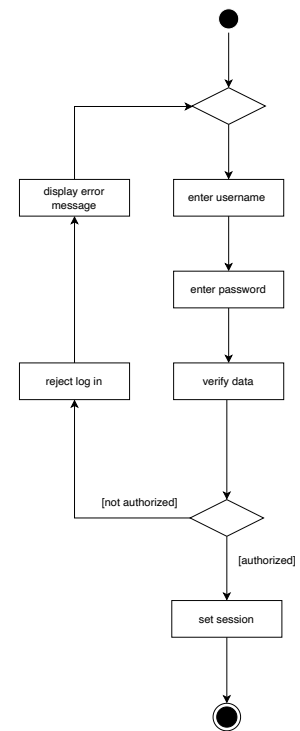
DAO:



3.2. Activity diagram

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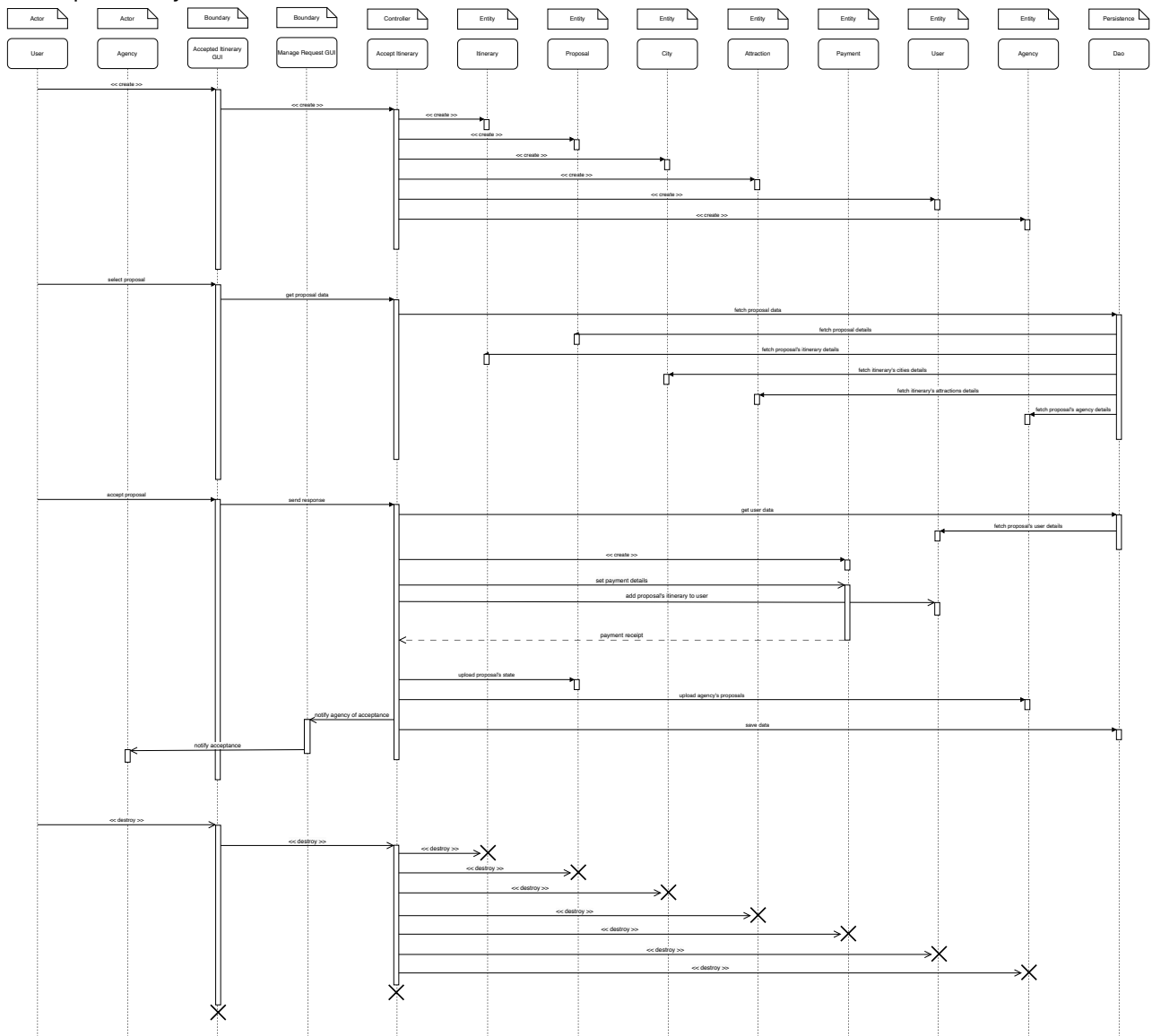


Simone Tummolo | 0309116:**Request Itinerary****Log In**

3.3. Sequence diagram

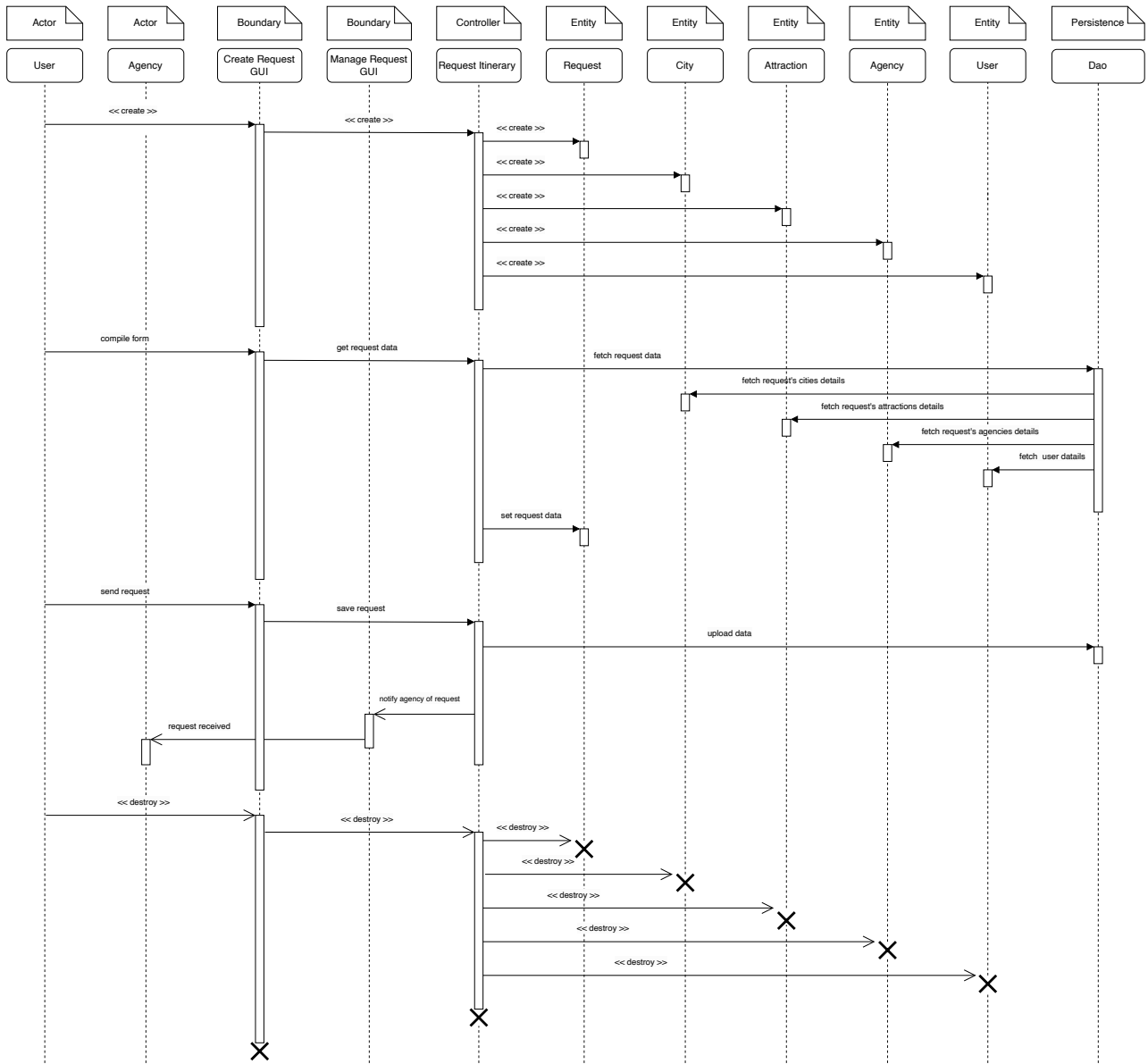
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Accept Itinerary Use Case:



Simone Tummolo | 0309116:**Request Itinerary Use Case:**

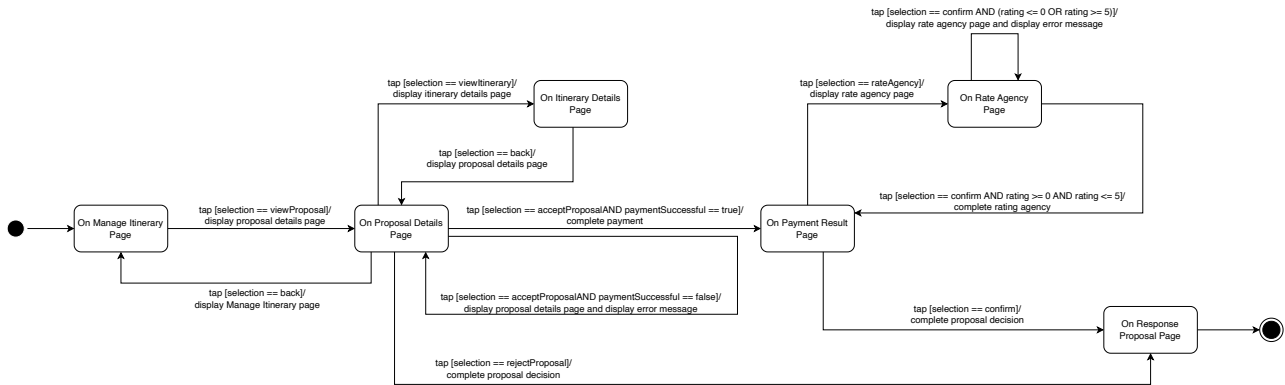
request itinerary



3.4. State diagram

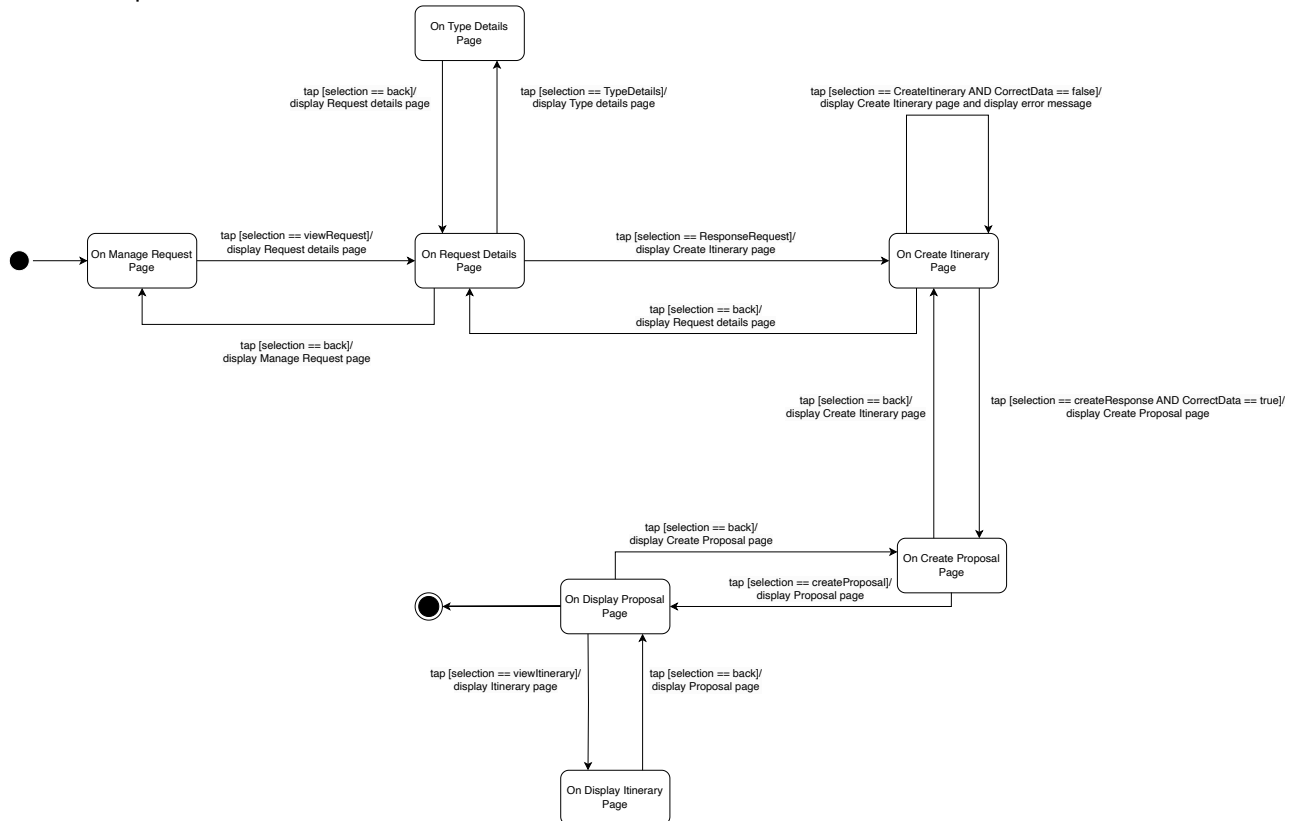
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Accept Itinerary Use Case:



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Send Response Use Case:



4. Testing

The test classes we have analysed can be found at the following [link](#).

In particular, we tested some methods of classes:

- **Teo Miozzi | 0309010:**
 - ItineraryBean → setDuration(), setOutboundFlightDepartureTime() and setAccommodations().
 - ProposalBean → setPrice(), setAccepted() and setDescription().
 - AcceptItineraryController → sendResponse(), getProposalItinerary() and getAccountType.
 - ManageItineraryController → getUserProposals().
- **Simone Tummolo | 0309116:**
 - RequestBean → setAccepted(), setDayNum() and setOtherRequests().
 - LoginController → login(), signup() and getUserType().
 - RequestItineraryController → createRequest(), getRequest() and saveRequest().

5. Code

The source code of the project is available on GitHub at the following link: [GitHub Repository](#).

The code quality analysis is available on SonarCloud at the following link: [SonarCloud Analysis](#).

6. Video

A video demonstration showing how the application works is available on YouTube at the following link: [YouTube video](#).